1NF

- This given table is in first normal form because the values are atomic and there are no repeating groups
- It has a composite primary key since CustomerNumber, orderDate, orderTime, and partNumber alone are not unique keys (because one customer can place many orders on different days and at different times and one order can have many parts) but together they make up a unique key. The customerNumber, orderDate, and orderTime identify one order and the part number specifies the line within it. So, these four together make up the composite primary key.

Happy Supplies Parts Warehouse					
Customer Name: Jeff Peterson			Date: 7/1/2024		
Customer Number: HG54587			Time: 10:30 am		
Customer Type: Consumer			Employees: D. Harrison		
partNumber	name	type	cageCode	quantityOrdered	unitPrice
10654	Float Control	Plumbing	G413	4	12
10456	Modulator	Electrical	H433	3	7
10776	Hose Assembly	Plumbing	G413	7	9
10657	Float Assembly	Plumbing	G413	5	10

2NF:

- The given table was not in second normal form since it has partial dependency because customerName and customerType depend only on customerNumber and not orderDate, orderTime, and partNumber, employee depends on customerNumber, orderDate, and orderTime and not on partNumber and part depends on partNumber and not customerNumber, orderDate, and orderTime.
- To bring the table in second normal form we break it into four tables to remove the partial dependency.
  - Since quantityOrdered and unitPrice depends on all four, that is, customerNumber, orderDate, orderTime, and partNumber (because it is the quanity and price of each part in a particular order) it stays and customerNumber, orderDate, orderTime, and partNumber together become the composite primary key. All four of these are also the foreign keys.
  - Since partName, partType, and cageCode depends only on partNumber we make a separate table with these three where partNumber is the primary key.
  - Since employee depends only on customerNumber, orderDate, and orderTime (because one employee handles an entire order not just one part) we put those 4 in a different table where customerNumber, orderDate, and orderTime are the primary keys. In this table, customerNumber is also a foreign key since it connects to the customer table.
  - Since customerName and customerType depend only on customerNumber we move those three to a different table where customerNo is the primary key.

order_detail

customerNumber	***orderDate***	***orderTime***	***partNumber***	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30 am	10654	4	12
HG54587	7/1/2024	10:30 am	10456	3	7
HG54587	7/1/2024	10:30 am	10776	7	9
HG54587	7/1/2024	10:30 am	10657	5	10

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

partNumber	partName	partType	cageCode
10654	Float Control	Plumbing	G413
10456	Modulator	Electrical	H433
10776	Hose Assembly	Plumbing	G413
10657	Float Assembly	Plumbing	G413

customer_order

customerNumber	**orderDate**	**orderTime**	Employee
HG54587	7/1/2024	10:30 am	D. Harrison

3NF:

- The given table is not in third normal form because it has transitive dependancies because the cage code only depends on partType (a non-key attribute) and not partNumber because parts of similar type are stored in one cage.
- To bring the table to third normal form we further break down the part table while everything else remains the same.
  - Since codeCage in the part table depends on partType we create a new event table where partType is the primary key. Additionally, partType also stays in the part table as a foreign key.

order_detail

customerNumber	***orderDate***	***orderTime***	***partNumber***	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30 am	10654	4	12
HG54587	7/1/2024	10:30 am	10456	3	7
HG54587	7/1/2024	10:30 am	10776	7	9
HG54587	7/1/2024	10:30 am	10657	5	10

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

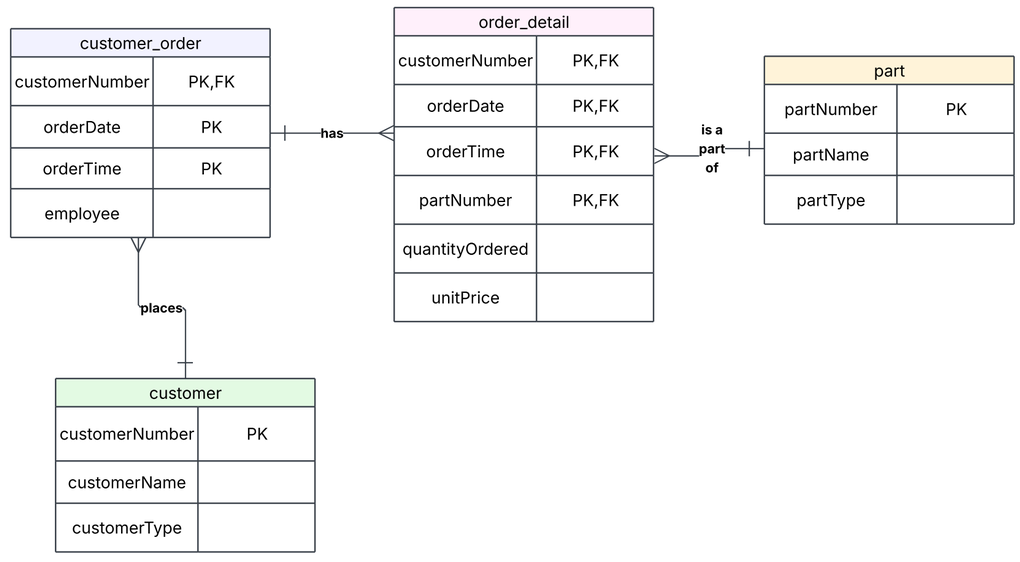
partNumber	partName	*partType*
10654	Float Control	Plumbing
10456	Modulator	Electrical
10776	Hose Assembly	Plumbing
10657	Float Assembly	Plumbing

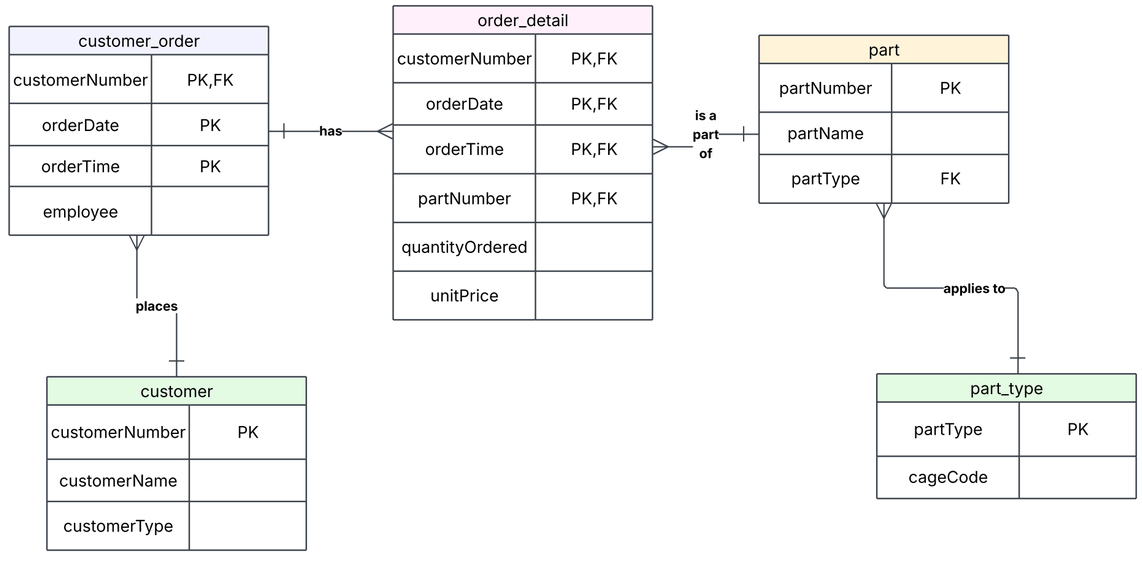
part_type

partType	cageCode
Plumbing	G413
Electrical	H433

customer_order

customerNumber	**orderDate**	**orderTime**	Employee
HG54587	7/1/2024	10:30 am	D. Harrison





Assumptions:

1. The form shows one order, but the table is designed for many orders. Each row is one part line of one order, and the order details are repeated on every row.
2. The customer number, order date and order time identiy one order, and the part number identifies a line within it. So the composite primany key is customerNumber+ orderDate+orderTime+partNumber.
3. One employee helps with a whole order and has no relationship with a specific customer, so employee depends on the order only (that is, customerNumber, orderDate, and orderTime). It holds a name, and the brief does not allow adding an employee number.
4. Each part type is stored in one cage, so cageCode is determined by partType.
5. The unit price is recorded for each order line, because the price of a part can change over time.

1NF

- This given table is in first normal form because the values are atomic and there are no repeating groups
- It has a composite primary key since staffNo, appointmentDate, and appointmentTime alone are not unique keys but together they make up a unique key. So, these three together make up the composite primary key

appointments

staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

2NF:

- The given table was not in second normal form since it has partial dependency because therapistName depends only on StaffNo and not the other two primary keys and branchNo depends only on staffNo and appointmentDate (since one therapist remains in one branch for any particular day).
- To bring the table in second normal form we break it into three tables to remove the partial dependency.
  - Since patNo and patName depend on all three, that is, staffNo, appointmentDate, and appointmentTime (since one patient's appointment depends on the therapist, the date, and the time of appointment) they stay and staffNo, appointmentDate, and appointmentTime become the composite primary key. staffNo and appointmentDate are also the foreign keys.
  - Since therapistName depends only on staffNo we make a separate table with the two where staffNo is the primary key.
  - Since branchNo depends only on staffNo and appointmentDate because each therapist is in one branch throughout a particular day so we put those three to a different table where staffNo and appointmentDate together make up the primary key.

appointment

staffNo	patNo	patName	***appointmentDate***	**appointmentTime**
S1011	P100	Lily White	9/12/2022	10:00
S1011	P105	Jill Baker	9/12/2022	12:00
S1024	P108	Andy McKee	9/12/2022	10:00
S1024	P108	Andy McKee	9/14/2022	14:00
S1032	P105	Jill Baker	9/14/2022	16:30
S1032	P110	Jimmy Winter	9/15/2022	18:00

3NF:

- The given table is not in third normal form because it has transitive dependency since patient name depends only on patient number(a non-key attribute) and not on the primary key.
- To bring the table to third normal form we further break down the appointment table while everything else remains the same.
  - Since patName in the appointment table depends on patNo we create a new event table where patNo is the primary key. Additionally, patNo also stays in the contract table as a foreign key.

appointment

staffNo	patNo	patName	***appointmentDate***	**appointmentTime**
S1011	P100	Lily White	9/12/2022	10:00
S1011	P105	Jill Baker	9/12/2022	12:00
S1024	P108	Andy McKee	9/12/2022	10:00
S1024	P108	Andy McKee	9/14/2022	14:00
S1032	P105	Jill Baker	9/14/2022	16:30
S1032	P110	Jimmy Winter	9/15/2022	18:00

patient

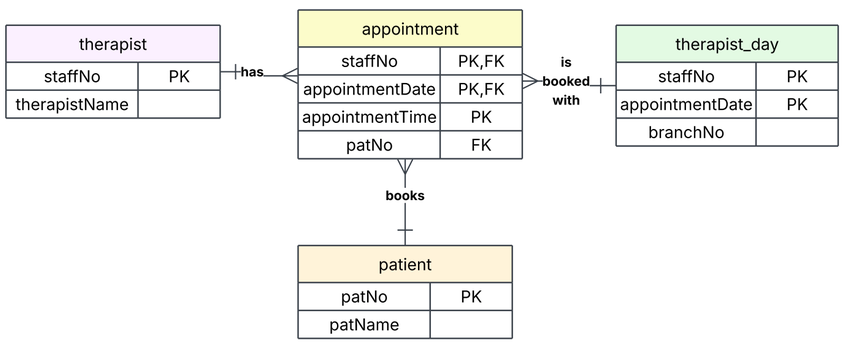
patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist_day

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

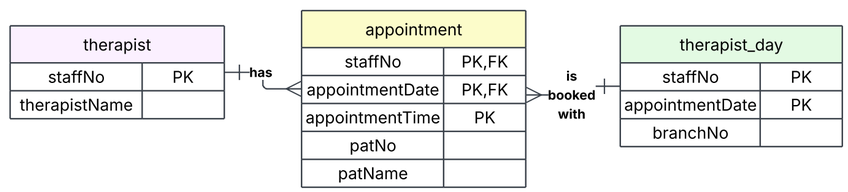


therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_day

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13



Assumptions:

1. Appointment time and appointment date are treated as separate atomic attributes.
2. Both the combination of staffNo, appointment date, and appointment time and the combination of patNo, appointment date, and appointment time are unique (because both therapist and patient can only be in one place at one time) but here we take staffNo+appointment date+ appointment time as the composite primary key because the question focuses more on the therapists.
3. A therapist sees patients at only one branch on a given day, so branchNo is determined by staffNo and apointmentDate.
4. StaffNo uniquely identifies a therapist and determines therapistName and patNo uniquely identifies a patient and determines patientName.

1NF

- This given table is in first normal form because the values are atomic and there are no repeating groups
- It has a composite primary key since eNo alone or contractNo alone are not unique keys but together they make up a unique key. So, this pair is the primary key

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

3NF:

- The given table is not in third normal form because it has transitive dependencies because each contract applied to one event, so contractNo determines eventNo but each event has one location so eventLoc depends on eventNo (a non-key attribute).
- To bring the table to third normal form we further break down the contract table while everything else remains the same.
  - Since eventLoc in the contract table depends on eventNo we create a new event table where eventNo is the primary key. Additionally, eventNo also stays in the contract table as a foreign key.

contract

contractNo	eventNo
C1024	H25
C1025	H4
C1026	H25

employee_contract

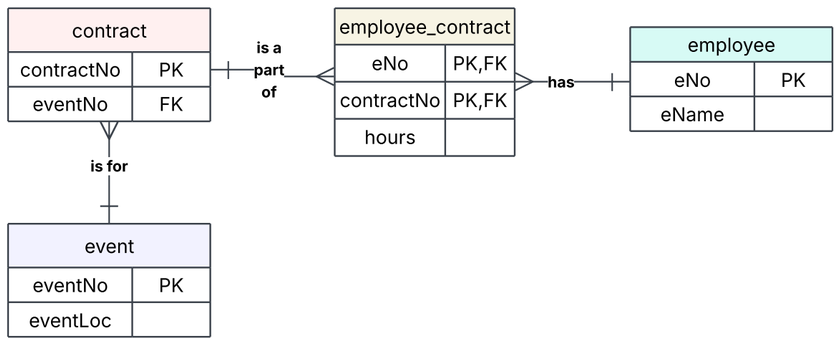
eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event

eventNo	eventLoc
H25	Queens
H4	Yonkers



2NF:

- The given table was not in second normal form since it has partial dependency because eName depends only on eNo and not on contractNo and eventNo and eventLoc depend only on the contractNo and not on eNo.
- To bring the table in second normal form we break it into three tables to remove the partial dependency.
  - Since hours depends on both eNo and contractNo (since it's the hours spent by one employee on one particular contract) it stays and eNo and contractNo become the composite primary key. Both eNo and contractNo are also the foreign keys.
  - Since eName depends only on eNo we make a separate table with the two where eNo is the primary key.
  - Since eventNo and eventLoc depend only on contractNo since each contract applies to one event we move those three to a different table where contractNo is the primary key.

contract

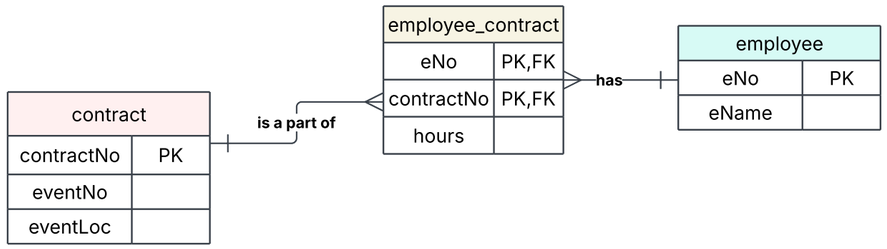
contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

employee_contract

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T



Assumptions:

1. The given table is named employee_contract. Every value is atomic, and each row records the hours one employee worked on one contract.
2. An employee can work on many contracts, and a contract can involve many employees, so neither eNo nor contractNo is unique on its own. Together they form the composite primary key.
3. eNo uniquely identifies an employee, as the brief states, so each eNo has exactly one eName.
4. contractNo uniquely identifies a contract. Each contract applied to only one event, as the brief states, so contractNo determines eventNo. An event can have several contracts
5. eventNo uniquely identifies an event, and each event has exactly one location, so eventNo determines eventLoc.